Essential ideas

8.1 Many reactions involve the transfer of a proton from an acid to a base.

8.2 The characterization of an acid depends on empirical evidence such as the production of gases in reactions with metals, the colour changes of indicators, or the release of heat in reactions with metal oxides and hydroxides.

8.3 The pH scale is an artificial scale used to distinguish between acid, neutral, and basic/alkaline solutions.

8.4 The pH depends on the concentration of the solution. The strength of acids or bases depends on the extent to which they dissociate in aqueous solution.

18.1 The acid–base concept can be extended to reactions that do not involve proton transfer.

18.2 The equilibrium law can be applied to acid–base reactions. Numerical problems can be simplified by making assumptions about the relative concentrations of the species involved. The use of logarithms is also significant here.

18.3 pH curves can be investigated experimentally but are mathematically determined by the dissociation constants of the acid and base. An indicator with an appropriate end point can be used to determine the equivalence point of the reaction.

8.5 Increased industrialization has led to greater production of nitrogen and sulfur oxides leading to acid rain, which is damaging our environment. These problems can be reduced through collaboration with national and intergovernmental organizations.

Acid–base theory informs much of our research into rainwater pollution and other aspects of environmental change.

The burning feeling of acid indigestion, the sour taste of grapefruit, and the vinegary smell of wine that has been exposed to the air are just some of the everyday encounters we have with acids. Likewise alkalis, or bases, are familiar substances – for example in baking soda, in household cleaners that contain ammonia, and in medication against indigestion. So what are the defining properties of these two groups of substances?

This question has intrigued chemists for centuries. The word 'acid' is derived from the Latin word *acetum* meaning sour – early tests to determine whether a substance was acidic were based on tasting! But it was learned that acids had other properties in common too: for example they changed the colour of the dye litmus from blue to red and they corroded metals. Similarly, alkalis were known to have distinctive properties such as being slippery to the touch, being able to remove fats and oils from fabrics, and turning litmus from red to blue. The name alkali comes from the Arabic word for plant ash, *alkalja*, where they were first identified. Early theories about acids and alkalis focused only on how they reacted together – it was actually suggested that the sourness of acids came from their possession of sharp angular spikes which became embedded in soft, rounded particles of alkali!

Over the last 120 years, our interpretation of acid–base behaviour has evolved alongside an increasing knowledge of atomic structure and bonding. In this chapter, we explore the modern definitions of acids and bases and learn how these help us to interpret and predict their interactions. Acid–base theory is central to topics such as air and water pollution, how global warming may affect the chemistry of the oceans, the action of drugs in the body, and many other aspects of cutting-edge research. As most acid–base reactions involve equilibria, much of the approach and mathematical content is based on work covered in Chapter 7. It is strongly recommended that you are familiar with this work before starting this chapter.

A food scientist testing a sample of a batch of orange juice in a factory in France. pH measurements are an important part of the quality control of the product.

8.1 Theories of acids and bases

Understandings:

- A Brønsted–Lowry acid is a proton/H^+ donor and a Brønsted–Lowry base is a proton/H^+ acceptor.

  Guidance
  Students should know the representation of a proton in aqueous solution as both $H^+(aq)$ and $H_3O^+(aq)$.
- Amphiprotic species can act as both Brønsted–Lowry acids and bases.
- A pair of species differing by a single proton is called a conjugate acid–base pair.

Applications and skills:

- Deduction of the Brønsted–Lowry acid and base in a chemical reaction.

  Guidance
  The location of the proton transferred should be clearly indicated. For example, CH_3COOH/CH_3COO^- rather than $C_2H_4O_2/C_2H_3O_2^-$.
- Deduction of the conjugate acid or conjugate base in a chemical reaction.

  Guidance
  Lewis theory is not required here.

Early theories

The famous French chemist Lavoisier proposed in 1777 that oxygen was the 'universal acidifying principle'. He believed that an acid could be defined as a compound of oxygen and a non-metal. In fact, the name he gave to the newly discovered gas *oxygen* means 'acid-former'. This theory, however, had to be dismissed when the acid HCl was proven to be made of hydrogen and chlorine only – no oxygen. To hold true, of course, any definition of an acid has to be valid for *all* acids.

A big step forward came in 1887 when the Swedish chemist Arrhenius suggested that an acid could be defined as a substance that dissociates in water to form hydrogen ions (H^+) and anions, while a base dissociates into hydroxide (OH^-) ions and cations. He also recognized that the hydrogen and hydroxide ions could form water, and the cations and anions form a salt. In a sense, Arrhenius was very close to the theory that is widely used to explain acid and base properties today, but his focus was only on aqueous systems. A broader theory was needed to account for reactions occurring without water, and especially for the fact that some insoluble substances show base properties.

Svante August Arrhenius (1859–1927) wrote up his ideas on acids dissociating into ions in water as part of his doctoral thesis while a student at Stockholm University. But his theory was not well received and he was awarded the lowest possible class of degree. Later, his work gradually gained recognition and he received one of the earliest Nobel Prizes in Chemistry in 1903.

Arrhenius may be less well known as the first person documented to predict the possibility of global warming as a result of human activity. In 1896, aware of the rising levels of CO_2 caused by increased industrialization, he calculated the likely effect of this on the temperature of the Earth. Today, over 100 years later, the significance of this relationship between increasing CO_2 and global temperatures has become a subject of major international concern.

NATURE OF SCIENCE

The evolution of theories to explain acid–base chemistry and develop general principles is a fascinating tale of the scientific process in action. Some theories have arisen and been disproved, such as Lavoisier's early definition of an acid as a compound containing oxygen. Falsification of an idea when it cannot be applied in all cases is an essential aspect of the scientific process. Other theories have proved to be too limited in application, such as Arrhenius' theory which could not be generalized beyond aqueous solutions.

On the other hand, Brønsted–Lowry theory (as well as an alternate broader theory known as Lewis theory that is described later), has stood the test of time and experimentation. This indicates that it has led to testable predictions which have supported the theory, and enabled wide ranging applications to be made.

Brønsted–Lowry: a theory of proton transfer

In 1923 two chemists, Martin Lowry of Cambridge, England, and Johannes Brønsted of Copenhagen, Denmark, working independently, published similar conclusions regarding the definitions of acids and bases. Their findings overcame the limitations of Arrhenius' work and have become established as the **Brønsted–Lowry theory**.

This theory focuses on the transfer of H^+ ions during an acid–base reaction: acids donate H^+ while bases accept H^+. For example, in the reaction between HCl and NH_3:

$$HCl + NH_3 \rightleftharpoons NH_4^+ + Cl^-$$

HCl transfers H^+ to NH_3 and so acts as an acid; NH_3 accepts the H^+ and so acts as a base.

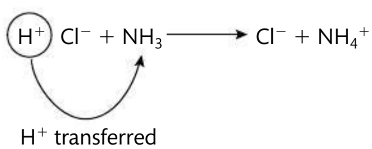

Figure 8.1 HCl transfers H^+ to NH_3.

Hydrogen atoms contain just one proton and one electron, so when they ionize by losing the electron, all that is left is the proton. Therefore *H^+ is equivalent to a proton*, and we will use the two terms interchangeably here.

The Brønsted–Lowry theory can therefore be stated as:

- a Brønsted–Lowry acid is a proton (H^+) donor;
- a Brønsted–Lowry base is a proton (H^+) acceptor.

Reaction between vapours of HCl and NH_3 forming the white smoke of ammonium chloride NH_4Cl.

$HCl(aq) + NH_3(aq) \rightarrow NH_4Cl(s)$

NATURE OF SCIENCE

Brønsted and Lowry's work on acid–base theory may be a good example of what is sometimes referred to as 'multiple independent discovery'. This refers to cases where similar discoveries are made by scientists at almost the same time, even though they have worked independently from each other. Examples include the independent discovery of the calculus by Newton and Leibniz, the formulation of the mechanism for biological evolution by Darwin and Wallace, and derivation of the pressure–volume relationships of gases by Boyle and Mariotte. Nobel Prizes in Chemistry are commonly awarded to more than one person working in the same field, who may have made the same discovery independently. Multiple independent discoveries are likely increasing as a result of communication technology, which enables scientists who may be widely separated geographically to have access to a common body of knowledge as the basis for their research.

When a discovery is made by more than one scientist at about the same time, ethical questions can arise, such as whether the credit belongs to one individual or should be shared, and whether there are issues regarding intellectual property and patent rights. Historically, these issues have been settled in very different ways in different cases. Nonetheless, the peer review process in science aims to ensure that credit is correctly awarded to the scientist or scientists responsible for a discovery, and that published work represents a new contribution to work in that field.

A Brønsted–Lowry acid is a proton (H^+) donor.

A Brønsted–Lowry base is a proton (H^+) acceptor.

Conjugate pairs

The act of donating cannot happen in isolation – there must always be something present to play the role of acceptor. In Brønsted–Lowry theory, an acid can therefore only behave as a proton donor if there is also a base present to accept the proton.

Let's consider the acid–base reaction between a generic acid HA and base B:

$$HA + B \rightleftharpoons A^- + BH^+$$

We can see that HA acts as an acid, donating a proton to B while B acts as a base, accepting the proton from HA. But if we look also at the reverse reaction, we can pick out another acid–base reaction: here BH^+ is acting as an acid, donating its proton to A^- while A^- acts as a base accepting the proton from BH^+. In other words acid HA has reacted to form the base A^-, while base B has reacted to form acid BH^+.

conjugate acid–base pair

$$HA + B \rightleftharpoons A^- + BH^+$$

conjugate acid–base pair

Acids react to form bases and vice versa. The acid–base pairs related to each other in this way are called **conjugate acid–base pairs**, and you can see that they *differ by just one proton*. It is important to be able to recognize these pairs in a Brønsted–Lowry acid–base reaction.

Note that $H_3O^+(aq)$ and $H^+(aq)$ are both used to represent a proton in aqueous solution.

One example of a conjugate pair is H_2O and H_3O^+, which is found in all acid–base reactions in aqueous solution. The reaction

$$H_2O(l) + H^+(aq) \rightleftharpoons H_3O^+(aq)$$

Lowry described the ready hydration of the proton as 'the extreme reluctance of the hydrogen nucleus to lead an isolated existence'.

occurs when a proton released from an acid readily associates with H_2O molecules forming H_3O^+. In other words, protons become hydrated. H_3O^+ is variously called the hydroxonium ion, the oxonium ion, or the hydronium ion and is always the form of hydrogen ions in aqueous solution. However, for most reactions it is convenient simply to write it as $H^+(aq)$. Note that in this pair H_3O^+ is the conjugate acid and H_2O its conjugate base.

Worked example

Label the conjugate acid–base pairs in the following reaction:

$$CH_3COOH(aq) + H_2O(l) \rightleftharpoons CH_3COO^-(aq) + H_3O^+(aq)$$

Solution

$$\underbrace{\underset{\text{acid}}{CH_3COOH} \,/\, \underset{\text{base}}{CH_3COO^-}}_{\text{conjugate pair}} \qquad \underbrace{\underset{\text{base}}{H_2O} \,/\, \underset{\text{acid}}{H_3O^+}}_{\text{conjugate pair}}$$

The fact that in a conjugate pair the acid always has one proton more than its conjugate base, makes it easy to predict the formula of the corresponding conjugate for any given acid or base.

Worked example

1 Write the conjugate base for each of the following.

(a) H_3O^+ (b) NH_3 (c) H_2CO_3

2 Write the conjugate acid for each of the following.

(a) NO_2^- (b) OH^- (c) CO_3^{2-}

Solution

1 To form the base from these species, remove one H^+

(a) H_2O (b) NH_2^- (c) HCO_3^-

2 To form the acid from these species, add one H^+

(a) HNO_2 (b) H_2O (c) HCO_3^-

Some species can act as acids and as bases

You may be surprised to see water described in the answers to the Worked examples above as a base (Q1, part a), and as an acid (Q2, part b), as you are probably not used to thinking of water as an acid, or as a base, but rather as a neutral substance. The point is that Brønsted–Lowry theory describes acids and bases in terms of how they react together, so it all depends on what water is reacting with. Consider the following:

$$\underset{\text{acid}}{CH_3COOH} + \underset{\text{base}}{H_2O} \rightleftharpoons \underset{\text{base}}{CH_3COO^-} + \underset{\text{acid}}{H_3O^+}$$

$$\underset{\text{base}}{NH_3} + \underset{\text{acid}}{H_2O} \rightleftharpoons \underset{\text{acid}}{NH_4^+} + \underset{\text{base}}{OH^-}$$

So with CH_3COOH, water acts as a Brønsted–Lowry base, but with NH_3 it acts as a Brønsted–Lowry acid.

Notice that water is not the only species that can act as both an acid and a base – for example, we see in the Worked example below that HCO_3^- behaves similarly. Substances which can act as Brønsted–Lowry acids and bases in this way are said to be **amphiprotic**. What are the features that enable these species to have this 'double identity'?

When writing the conjugate acid of a base, add one H^+; when writing the conjugate base of an acid, remove one H^+. Remember to adjust the charge by the 1+ removed or added.

The acid and base in a conjugate acid–base pair differ by just one proton.

Amphoteros is a Greek word meaning 'both'. For example amphibians are adapted both to water and to land.

An amphiprotic substance is one which can act as both a proton donor and a proton acceptor.

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Terminology in science has to be used appropriately according to the context. *Amphiprotic* specifically relates to Brønsted–Lowry acid–base theory, where the emphasis is on the transfer of a proton. The term *amphoteric*, on the other hand, has a broader meaning as it is used to decribe a substance which can act as an acid and as a base, including reactions that do not involve the transfer of a proton. For example, as described in Chapter 3, aluminium oxide is an amphoteric oxide as it reacts with both dilute acids and alkalis. We will see in section 18.1 that this acid–base behaviour is best described by a different theory, the Lewis theory. Note that all amphiprotic substances are also amphoteric, but the converse is not true. The fact that the two terms exist reflects the fact that different theories are used to describe acid–base reactions.

To act as a Brønsted–Lowry acid, they must be able to dissociate and release H^+.

To act as a Brønsted–Lowry base, they must be able to accept H^+, which means they must have a lone pair of electrons.

So substances that are amphiprotic according to Brønsted–Lowry theory must possess both a lone pair of electrons and hydrogen that can be released as H^+.

Worked example

Write equations to show HCO_3^- acting (a) as a Brønsted–Lowry acid and (b) as a Brønsted–Lowry base.

Solution

(a) to act as an acid, it donates H^+

$$HCO_3^-(aq) + H_2O(l) \rightleftharpoons CO_3^{2-}(aq) + H_3O^+(l)$$

(b) to act as a base, it accepts H^+

$$HCO_3^-(aq) + H_2O(l) \rightleftharpoons H_2CO_3(aq) + OH^-(aq)$$

Exercises

1 Deduce the formula of the conjugate acid of the following:

(a) SO_3^{2-} **(b)** CH_3NH_2 **(c)** $C_2H_5COO^-$ **(d)** NO_3^- **(e)** F^- **(f)** HSO_4^-

2 Deduce the formula of the conjugate base of the following:

(a) H_3PO_4 **(b)** CH_3COOH **(c)** H_2SO_3 **(d)** HSO_4^- **(e)** OH^- **(f)** HBr

3 For each of the following reactions, identify the Brønsted–Lowry acids and bases and the conjugate acid–base pairs:

(a) $CH_3COOH + NH_3 \rightleftharpoons NH_4^+ + CH_3COO^-$

(b) $CO_3^{2-} + H_3O^+ \rightleftharpoons H_2O + HCO_3^-$

(c) $NH_4^+ + NO_2^- \rightleftharpoons HNO_2 + NH_3$

4 Show by means of equations how the anion in K_2HPO_4 is amphiprotic.

8.2 Properties of acids and bases

Understandings:

- Most acids have observable characteristic chemical reactions with reactive metals, metal oxides, metal hydroxides, hydrogen carbonates, and carbonates.

  Guidance
  Bases which are not hydroxides, such as ammonia, soluble carbonates, and hydrogen carbonates should be covered.
- Salt and water are produced in exothermic neutralization reactions.

Applications and skills:

- Balancing chemical equations for the reaction of acids.
- Identification of the acid and base needed to make different salts.
- Candidates should have experience of acid–base titrations with different indicators.

  Guidance
  The colour changes of different indicators are given in the data booklet in section 22.

While we have commented that ideas regarding the defining nature of acids and bases have been long debated, the recognition of what these substances *do* has been known for centuries.

We will look here at some typical reactions of acids and bases in aqueous solutions where H^+ is the ion common to all acids. The bases considered here are those that neutralize acids to produce water, which include:

- metal oxides and hydroxides;
- ammonia;
- soluble carbonates (Na_2CO_3 and K_2CO_3) and hydrogencarbonates ($NaHCO_3$ and $KHCO_3$).

The soluble bases are known as **alkalis**. When dissolved in water they all release the hydroxide ion OH^-. For example:

$$K_2O(s) + H_2O(l) \rightarrow 2K^+(aq) + 2OH^-(aq)$$

$$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$$

$$CO_3^{2-}(aq) + H_2O(l) \rightleftharpoons HCO_3^-(aq) + OH^-(aq)$$

$$HCO_3^-(aq) \rightleftharpoons CO_2(g) + OH^-(aq)$$

Acids react with metals, bases, and carbonates to form salts

The term **salt** refers to the ionic compound formed when the hydrogen of an acid is replaced by a metal or another positive ion. Salts form by reaction of acids with metals or bases. The familiar example, sodium chloride, NaCl, known as common salt, is derived from the acid HCl by reaction with a base that contains Na, such as NaOH. The terms **parent acid** and **parent base** are sometimes used to describe this relationship between an acid, a base, and their salt.

parent base NaOH → H_2O ← HCl parent acid

NaOH → NaCl ← HCl

salt

There are three main types of reaction by which acids react to form salts.

1 Acid + metal → salt + hydrogen

$$2HCl(aq) + Zn(s) \rightarrow ZnCl_2(aq) + H_2(g)$$

$$H_2SO_4(aq) + Fe(s) \rightarrow FeSO_4(aq) + H_2(g)$$

$$2CH_3COOH(aq) + Mg(s) \rightarrow Mg(CH_3COO)_2(aq) + H_2(g)$$

We can also write these as ionic equations. For example:

$$2H^+(aq) + \cancel{2Cl^-(aq)} + Zn(s) \rightarrow Zn^{2+}(aq) + \cancel{2Cl^-(aq)} + H_2(g)$$

NATURE OF SCIENCE

Classification leading to generalizations is an important aspect of studies in science. Historically, groups of substances such as acids were classified together because they were shown to have similar chemical properties. Over time it was recognized that many substances share these acidic properties, and so the classification broadened. Similarly, grouping of compounds as bases and alkalis occurred on the basis of experimental evidence for their properties.

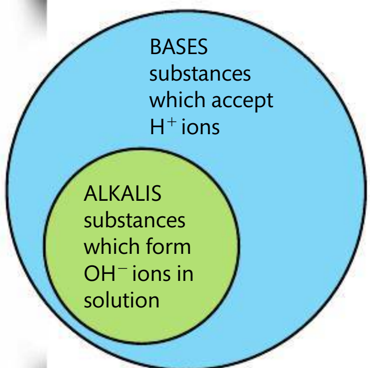


Figure 8.2 The relationship between alkalis and bases.

Alkalis are bases that dissolve in water to form the hydroxide ion OH^-.

In acid–base theory the words ionization and dissociation are often used interchangeably as acid dissociation always leads to ion formation.

Magnesium reacting with HCl.
$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$
The tiny bubbles are hydrogen gas being liberated.

Species which do not change during a reaction, like Cl^- here, are called **spectator ions** and can be cancelled out. So the net reaction is:

$$2H^+(aq) + Zn(s) \rightarrow Zn^{2+}(aq) + H_2(g)$$

These reactions of metals with acids are the reason why acids have corrosive properties on most metals, and why, for example, it is important to keep car battery acid well away from the metal bodywork of the car.

You can demonstrate the release of hydrogen from acids in simple experiments by adding a small piece of metal to a dilute solution of the acid. There is a big range, however, in the reactivity of metals in these reactions. More reactive metals such as sodium and potassium in Group 1 would react much too violently, while copper and other less reactive metals such as silver and gold will usually not react at all. This is partly why these less reactive metals are so valuable – they are much more resistant to corrosion. We will consider this differing reactivity in Chapter 9. Another point to note is that although the common acid nitric acid, HNO_3, does react with metals, it usually does not release hydrogen owing to its oxidizing properties (Chapter 9).

Acid + reactive metal → salt + hydrogen

2 Acid + base → salt + water

$$HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$$

$$HNO_3(aq) + NH_4OH(aq) \rightarrow NH_4NO_3(aq) + H_2O(l)$$

$$2CH_3COOH(aq) + CuO(s) \rightarrow Cu(CH_3COO)_2(aq) + H_2O(l)$$

These reactions between acids and bases are known as **neutralization** reactions. They can all be represented by one common ionic equation that shows the net reaction clearly:

$$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$$

Neutralization reactions of acids with bases are exothermic – heat is released. The **enthalpy of neutralization** is defined as the enthalpy change that occurs when an acid and a base react together to form one mole of water. For reactions between all strong acids and strong bases, the enthalpy change is very similar with $\Delta H = -57$ kJ mol^{-1} approximately. This is because the net reaction is the same, involving the formation of water from its ions.

There are times when neutralization reactions are useful to help reduce the effect of an acid or a base. For example, treatment for acid indigestion often involves using 'antacids' which contain a mixture of weak alkalis such as magnesium hydroxide and aluminium hydroxide. As we will learn in section 8.5, in places where the soil has become too acidic, the growth of many plants will be restricted. Adding a weak alkali such as lime, CaO, can help to reduce the acidity and so increase the fertility of the soil.

Bee stings are slightly acidic, and so have traditionally been treated by using a mild alkali such as baking soda ($NaHCO_3$). Wasp stings, on the other hand, are claimed to be alkali, and so are often treated with the weak acid, ethanoic acid (CH_3COOH) found in vinegar. Whether these claims are valid is, however, open to dispute as the pH of wasp stings is actually very close to neutral. Nonetheless the healing powers of vinegar are well documented and vigorously defended.

3 Acid + carbonate → salt + water + carbon dioxide

$$2HCl(aq) + CaCO_3(s) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$$

$$H_2SO_4(aq) + Na_2CO_3(aq) \rightarrow Na_2SO_4(aq) + H_2O(l) + CO_2(g)$$

$$CH_3COOH(aq) + KHCO_3(aq) \rightarrow KCH_3COO(aq) + H_2O(l) + CO_2(g)$$

These reactions can also be represented as an ionic equation:

$$2H^+(aq) + CO_3^{2-}(aq) \rightarrow H_2O(l) + CO_2(g)$$

Acid + carbonate → salt + water + carbon dioxide

Neutralization reactions occur when an acid and base react together to form a salt and water. They are exothermic reactions.

The reactions, like the reaction of acids with metals, involve a gas being given off so they visibly produce bubbles, known as **effervescence**.

Rain water dissolves some carbon dioxide from the air to form a weak solution of carbonic acid, H_2CO_3.

$$H_2O(l) + CO_2(g) \rightleftharpoons H_2CO_3(aq)$$

Greater pressure, such as that found in capillary beds of limestone ($CaCO_3$) rocks, increases the tendency of CO_2 to dissolve, giving rise to a more acidified solution. This then reacts on the limestone as follows:

$$H_2CO_3(aq) + CaCO_3(s) \rightleftharpoons Ca(HCO_3)_2(aq)$$

$CaCO_3$, limestone, is virtually insoluble whereas the product calcium hydrogencarbonate, $Ca(HCO_3)_2$, is soluble and washes away, leading to erosion of the rocks. This is why caves commonly form in limestone regions. Inside the cave where the pressure is lower, the reaction above may be reversed, as less CO_2 dissolves. In this case $CaCO_3$ comes out of solution and precipitates, giving rise to the formations known as stalagtites and stalagmites.

Similar reactions of rain water dissolving $CaCO_3$ rocks can give rise to water supplies with elevated levels of Ca^{2+} ions, known as 'hard water'.

Kitchen chemistry. Baking soda ($NaHCO_3$) and vinegar (CH_3COOH) react together in a powerful acid–base reaction, releasing carbon dioxide gas.

$$NaHCO_3(s) + CH_3COOH(aq) \rightarrow NaCH_3COO(aq) + H_2O(l) + CO_2(g)$$

In section 8.5, these reactions between acids and metals and metal carbonates are discussed in the context of the environmental problem known as acid deposition.

Acids and bases can be distinguished using indicators

Indicators act as chemical detectors, giving information about a change in the environment. The indicators most widely used in chemistry are **acid–base indicators** that change colour reversibly according to the concentration of H^+ ions in the solution. This happens because they are weak acids and bases whose conjugates have different colours. The colour change means that they can be used to identify the pH of a substance. Indicators are generally used either as aqueous solutions or absorbed onto 'test paper'.

Probably the best known acid–base indicator is **litmus**, which is a dye derived from lichens, and which turns pink in the presence of acid and blue in the presence of alkalis. It is widely used to test for acids or alkalis, but is not so useful in distinguishing between different strengths of acid or alkali.

Other indicators give different colours in different solutions of acid and alkali. Some common examples are given in the table here, and there are more in section 22 of your IB data booklet.

Limestone rock shaped by natural chemical erosion in Switzerland. Rainwater, a weak solution of H_2CO_3, reacts with the calcium carbonate, slowly dissolving it.

Indicator	Colour in acid	Colour in alkali
litmus	pink	blue
methyl orange	red	yellow
phenolphthalein	colourless	pink

Many of these indicators are derived from natural substances such as extracts from flower petals and berries. A common indicator in the laboratory is **universal indicator**, which is formed by mixing together several indicators. It changes colour

Litmus indicator compared in acidic and alkaline solutions.

Be careful not to assume that indicator tests always show acids as pink and alkalis as blue. This is true with litmus, but other indicators give many different colours including pink in alkali.

many times across a range of different acids and alkalis and so can be used to measure the concentration of H^+ on the pH scale. We will discuss this later in the chapter.

In section 18.3 (page 378), we will learn in more detail how indicators work and how they are used in quantitative experimental work.

Acid–base titrations are based on neutralization reactions

Neutralization reactions are often used in the laboratory to calculate the exact concentration of an acid or an alkali when the other concentration is known. As we learned in Chapter 1, the solution of known concentration is known as the standard solution. The technique known as **acid–base titration** involves reacting together a carefully measured volume of one of the solutions, and adding the other solution gradually until the so-called **equivalence point** is reached where they exactly neutralize each other. A convenient way to determine when the equivalence point has been reached is to use an indicator, chosen to change colour as the acid and base exactly neutralize each other.

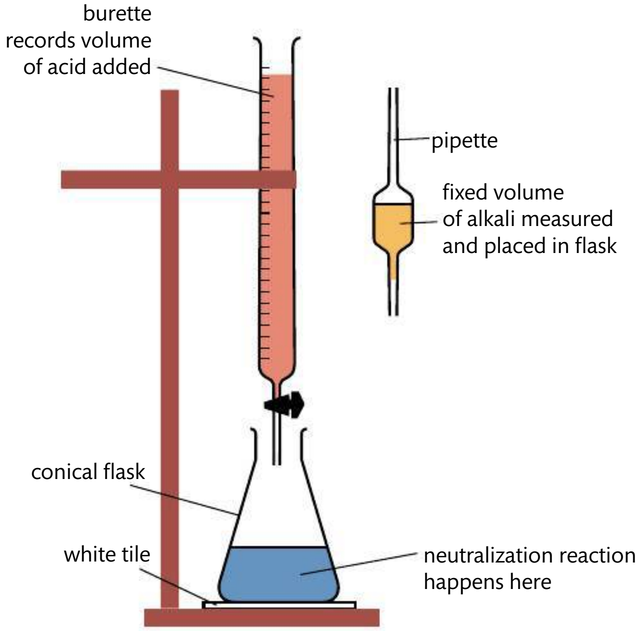


Figure 8.3 Simple titration apparatus.

For example, titration can be used in the following experiments:

- to calculate the concentration of ethanoic acid in vinegar by titration with a standard solution of aqueous sodium hydroxide, using phenolphthalein indicator;
- to calculate the concentration of sodium hydroxide by titration with a standard solution of hydrochloric acid, using methyl orange indicator.

A good indicator is one that gives a distinct or sharp colour change at the equivalence point.

In section 18.3 we will study additional examples of titrations and learn how to choose an indicator that is appropriate for specific combinations of acid and base.

Acid–base titration to calculate the concentration of ethanoic acid in vinegar

Full details of how to carry out this experiment with a worksheet are available online.

Exercises

5 Write equations for the following reactions:

(a) sulfuric acid and copper oxide
(b) nitric acid and sodium hydrogencarbonate
(c) phosphoric acid and potassium hydroxide
(d) ethanoic acid and aluminium

6 An aqueous solution of which of the following reacts with calcium metal?

A ammonia
B hydrogen chloride
C potassium hydroxide
D sodium hydrogencarbonate

7 Which of the following is / are formed when a metal oxide reacts with a dilute acid?

I a metal salt
II water
III carbon dioxide gas

A I only **B** I and II only **C** II and III only **D** I, II, and III

8 Suggest by name a parent acid and parent base that could be used to make the following salts. Write equations for each reaction.

(a) sodium nitrate
(b) ammonium chloride
(c) copper(II) sulfate
(d) potassium methanoate

When you need to choose a base in solution to make a specific salt, it is useful to know some simple solubility rules.

- The only soluble carbonates and hydrogencarbonates are $(NH_4)_2CO_3$, Na_2CO_3, K_2CO_3, $NaHCO_3$, $KHCO_3$, and $Ca(HCO_3)_2$.
- The only soluble hydroxides are NH_4OH, LiOH, NaOH, and KOH.

8.3 The pH scale

Understandings:

- pH = $-\log[H^+(aq)]$ and $[H^+] = 10^{-pH}$.
- A change of one pH unit represents a 10-fold change in the hydrogen ion concentration $[H^+]$.

Guidance
Knowing the temperature dependence of K_w is not required.

- pH values distinguish between acidic, neutral, and alkaline solutions.
- The ionic product constant, $K_w = [H^+][OH^-] = 10^{-14}$ at 298 K.

Applications and skills:

- Solving problems involving pH, $[H^+]$, and $[OH^-]$.

Guidance
- *Students should be concerned only with strong acids and bases in this sub-topic.*
- *Students will not be assessed on pOH values.*

- Students should be familiar with the use of a pH meter and universal indicator.

pH is a logarithmic expression of $[H^+]$

Chemists realized a long time ago that it would be useful to have a quantitative scale of acid strength based on the concentration of hydrogen ions. As the majority of acids encountered are weak, the hydrogen ion concentration expressed directly as mol dm^{-3} produces numbers with large negative exponents; for example the H^+ concentration in our blood is 4.6×10^{-8} mol dm^{-3}. Such numbers are not very user-friendly when it comes to describing and comparing acids. The introduction of the pH scale in 1909 by Sorensen led to wide acceptance owing to its ease of use. It is defined as follows:

$$pH = -\log_{10}[H^+]$$

The Danish chemist Søren Peder Lauritz Sorensen (1868–1939) developed the pH concept in 1909, originally proposing that it be formulated as p_H. He did not account for his choice of the letter 'p' though it has been suggested to originate from the German word *potenz* for power. It could equally well derive from the Latin, Danish, or French terms for the same word.

TOK

The pH scale can be considered to be an artificial or an arbitrary scale. To what extent is this true of all scales used in measuring?

$pH = -\log_{10}[H^+]$; $[H^+] = 10^{-pH}$

In other words, pH is the negative number to which the base 10 is raised to give the $[H^+]$. This can also be expressed as:

$$[H^+] = 10^{-pH}$$

- A solution that has $[H^+] = 0.1\ mol\ dm^{-3} \Rightarrow [H^+] = 10^{-1}\ mol\ dm^{-3} \Rightarrow pH = 1$.
- A solution that has $[H^+] = 0.01\ mol\ dm^{-3} \Rightarrow [H^+] = 10^{-2}\ mol\ dm^{-3} \Rightarrow pH = 2$.

Now we can explore some of the features of the pH scale that help to make it so convenient.

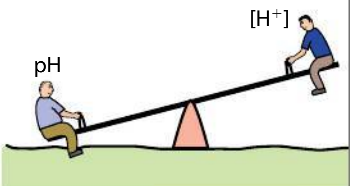


Figure 8.4 The inverse relationship between pH and H^+.

pH numbers are usually positive and have no units

Although the pH scale is theoretically an infinite scale (and can even extend into negative numbers), most acids and bases encountered will have positive pH values and fall within the range 0–14, corresponding to $[H^+]$ from 1.0 mol dm^{-3} to 10^{-14} mol dm^{-3}.

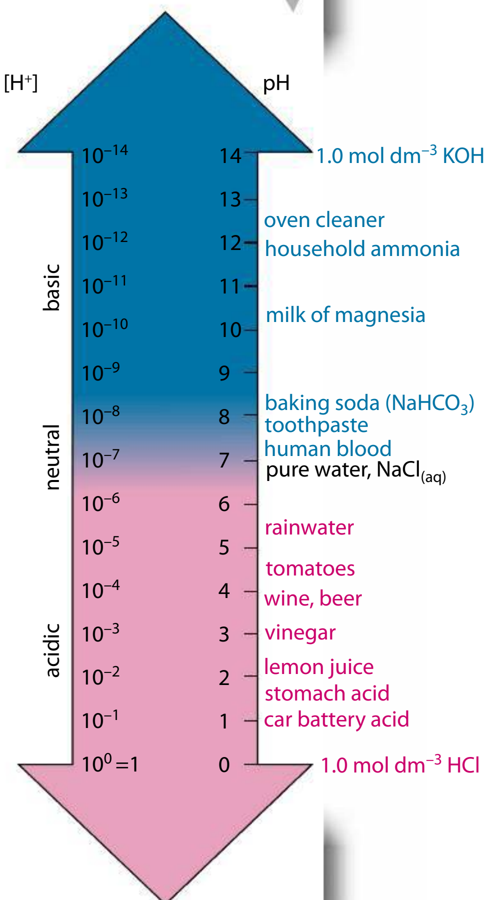
The pH number is inversely related to the $[H^+]$

Solutions with a higher $[H^+]$ have a lower pH and vice versa. So stronger and more concentrated acids have a lower pH, weaker and more dilute acids have a higher pH.

A change of one pH unit represents a 10-fold change in $[H^+]$

This means increasing the pH by one unit represents a decrease in $[H^+]$ by 10 times; decreasing by one pH unit represents an increase in $[H^+]$ by 10 times.

Figure 8.5 The pH scale at 298 K and pH values of some common substances.



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The use of a logarithmic scale to represent data enables a wide range of values to be presented as a smaller range of simpler numbers. We see here that the pH scale takes a range of hydrogen ion concentrations from 10^0 to 10^{-14} and effectively compresses it to a much smaller scale of numbers, 0–14. On page 88, we saw that the use of a logarithmic scale to represent the ionization energies of aluminium similarly makes a wide range of data easier to interpret. In cases of exponential change, a log base scale shows the data as a straight line, so outliers where the change is greater or less than exponential can be easily identified.

But use of a logarithmic scale can also be misleading, as it can obscure the scale of change if not interpreted correctly. For example, a small change in pH represents a dramatic difference in the hydrogen ion concentration of a solution. Keep this in mind when you read reports of changes in pH, for example of rainfall as a result of pollution. A reported change from pH 5.5 to pH 4.5 may not sound much, but in fact it represents a *ten-fold increase* in the hydrogen ion concentration – hugely significant to the acidic properties. As we will learn in section 18.3 (page 378), the pH of our blood is carefully controlled by chemicals called buffers to remain at 7.4; a change of only half a pH unit on either side of this is known to be fatal. Communication of data in science takes different forms and uses different scales, and must always be interpreted in the context used.

Logarithmic scales are used in other disciplines where they can help in the presentation of data. In medicine, the logarithmic decay of levels of drugs in the blood against time is used. In seismic studies, the Richter scale is a logarithmic expression of the relative energy released during earthquakes. In sense perception the decibel scale of sound intensity and the scale for measuring the visible brightness of stars (their magnitude) are also logarithmic scales.

Worked example

If the pH of a solution is changed from 3 to 5, deduce how the hydrogen ion concentration changes.

Solution

$pH = 3 \Rightarrow [H^+] = 10^{-3}$ mol dm^{-3}; $pH = 5 \Rightarrow [H^+] = 10^{-5}$ mol dm^{-3}

So $[H^+]$ has changed by 10^{-2} or decreased by 100.

The pH scale is a measure of $[H^+]$ and so at first glance may appear to be more suited for the measurement of acids than of bases. But we can in fact use the same scale to describe the alkalinity of a solution. As will be explained on page 359, this is because the relationship between $[H^+]$ and $[OH^-]$ is inverse in aqueous solutions, and so lower $[H^+]$ (higher pH) means higher $[OH^-]$ and vice versa. Therefore the scale of pH numbers represents a range of values from strongly acidic through to strongly alkaline.

pH 0	**pH 7**	**pH 14**
increasing $[H^+]$ **increasing acidity**	**neutral**	**increasing $[OH^-]$** **increasing alkalinity**

pH calculations

From the definition of pH we can:

(i) calculate the value of pH from a known concentration of H^+;

(ii) calculate the concentration of H^+ from a given pH.

Worked example

A sample of lake water was analysed at 298 K and found to have $[H^+] = 3.2 \times 10^{-5}$ mol dm^{-3}. Calculate the pH of this water and comment on its value.

Solution

$$pH = -\log_{10}[H^+]$$
$$= -\log_{10}(3.2 \times 10^{-5}) = -(-4.49485)$$
$$pH = 4.49$$

At 298 K this pH < 7, and the lake water is therefore acidic.

Worked example

Human blood has a pH of 7.40. Calculate the concentration of hydrogen ions present.

Solution

$$[H^+] = 10^{-pH}$$
$$= 10^{-7.40} = 4.0 \times 10^{-8}$$
$$[H^+] = 4.0 \times 10^{-8} \text{ mol dm}^{-3}$$

In the last 40 years, researchers have developed so-called 'super acids' by mixing together various substances. These are several orders of magnitude more acidic than conventional acids, and one example known as 'magic acid' is even able to dissolve candle wax.

In pH calculations you will need to be able to work out logarithms to base 10 and anti-logarithms. Make sure that you are familiar with these operations on your calculator.

Note that the logarithms used in all the work on acids and bases are logarithms to base 10 (log). Do not confuse these with natural logarithms, to base e, (ln).

The rule to follow for significant figures in logarithms is that *the number of decimal places in the logarithm should equal the number of significant figures in the number*. In other words, the number to the left of the decimal place in the logarithm is not counted as a significant figure (this is because it is derived from the exponential part of the number). In the first example here, the number given has two significant figures (3.2) and so the answer is given to two decimal places (.49).

Use of a pH meter and universal indicator in measuring pH

Full details of how to carry out this experiment with a worksheet are available online.

A digital probe showing a pH of 3.20 for the solution in the beaker. The temperature dial is adjusted to the room temperature, allowing the pH meter to compensate for the temperature.

Measuring pH

An easy way to measure pH is with universal indicator paper or solution. The substance tested will give a characteristic colour, which can then be compared with a colour chart supplied with the indicator. Narrower range indicators give a more accurate reading than broad range, but they always depend on the ability of the user's eyes to interpret the colour.

A more objective and usually more accurate means is by using a **pH meter** or probe that directly reads the $[H^+]$ concentration through a special electrode. pH meters can record to an accuracy of several decimal points. They must, however, be calibrated before each use with a buffer solution, and standardized for the temperature, as pH is a temperature-dependent measurement.

The pH scale and universal indicator. The tubes contain universal indicator added to solutions of pH 0–14 from left to right

The ionization of water

Victoria Falls on the Zimbabwe–Zambia border. It is estimated that 6.25×10^8 dm^3 of water flow over the falls every minute. According to the ionization constant discussed here, for every billion of these molecules only two are ionized, and that proportion gets even smaller at lower temperatures.

As the majority of acid–base reactions involve ionization in aqueous solution, it is useful to consider the role of water in more detail. Water itself does ionize, albeit only very slightly at normal temperatures and pressures, so we can write an equilibrium expression for this reaction.

$$H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$$

Therefore $K_c = \dfrac{[H^+][OH^-]}{[H_2O]}$

We are using $H^+(aq)$ throughout this chapter as a simplified form of $H_3O^+(aq)$. This is acceptable in most situations, but don't forget that H^+ in aqueous solution always exists as $H_3O^+(aq)$.

The concentration of water can be considered to be constant due to the fact that so little of it ionizes, and it can therefore be combined with K_c to produce a modified equilibrium constant known as K_w.

$K_c[H_2O] = [H^+][OH^-]$

Therefore $K_w = [H^+][OH^-]$

K_w is known as the **ionic product constant of water** and has a fixed value at a specified temperature. At 298 K, $K_w = 1.00 \times 10^{-14}$.

In pure water because $[H^+] = [OH^-]$, it follows that $[H^+] = \sqrt{K_w}$

So, at 298 K, $[H^+] = 1.00 \times 10^{-7}$ which gives pH = 7.00

This is consistent with the widely known value for the pH of water at room temperature.

Water is a pure liquid so its concentration is just a function of its density, which is 1.00 g cm^{-3}.

From $n = \frac{m}{M}$, this gives

$n(H_2O) = \frac{1000}{18}$ in 1 dm^3

so concentration (H_2O) = 55 mol dm^{-3}.

The relationship between H^+ and OH^- is inverse

Because the product $[H^+] \times [OH^-]$ gives a constant value, it follows that the concentrations of these ions must have an inverse relationship. In other words, in aqueous solutions the higher the concentration of H^+ the lower the concentration of OH^-. Solutions are defined as acidic, neutral, or basic according to their relative concentrations of these ions as shown below.

The value $K_w = 1.00 \times 10^{-14}$ at 298 K is given in section 2 of the IB data booklet so does not have to be learned.

		at 298 K
Acidic solutions are defined as those in which	**$[H^+] > [OH^-]$**	**pH < 7**
Neutral solutions are defined as those in which	**$[H^+] = [OH^-]$**	**pH = 7**
Alkaline solutions are defined as those in which	**$[H^+] < [OH^-]$**	**pH > 7**

$K_w = [H^+][OH^-]$

The concentrations of H^+ and OH^- are inversely proportional in an aqueous solution.

So if we know the concentration of either H^+ or OH^-, we can calculate the other from the value of K_w.

$$[H]^+ = \frac{K_w}{[OH]^-} \text{ and } [OH]^- = \frac{K_w}{[H]^+}$$

Worked example

A sample of blood at 298 K has $[H^+] = 4.60 \times 10^{-8}$ mol dm^{-3}.

Calculate the concentration of OH^- and state whether the blood is acidic, neutral, or basic.

Solution

At 298 K, $K_w = 1.00 \times 10^{-14} = [H^+][OH^-]$

$$[OH]^- = \frac{1.00 \times 10^{-14}}{4.60 \times 10^{-8}} = 2.17 \times 10^{-7} \text{ mol dm}^{-3}$$

As $[OH^-] > [H^+]$ the solution is basic.

Calculations involving $[H^+]$, $[OH^-]$, and pH, as well as the pOH scale, are considered in more detail in section 18.2, page 366.

Exercises

9 What happens to the pH of an acid when 10 cm^3 of it is added to 90 cm^3 of water?

10 Beer has a hydrogen ion concentration of 1.9×10^{-5} mol dm^{-3}. What is its pH?

11 An aqueous solution has a pH of 9 at 25 °C. What are its concentrations for H^+ and OH^-?

12 For each of the following aqueous solutions, calculate $[OH^-]$ from $[H^+]$ or $[H^+]$ from $[OH^-]$. Classify each solution as acidic, basic, or neutral at 298 K.

(a) $[H^+] = 3.4 \times 10^{-9}$ mol dm^{-3}
(b) $[OH^-] = 0.010$ mol dm^{-3}
(c) $[OH^-] = 1.0 \times 10^{-10}$ mol dm^{-3}
(d) $[H^+] = 8.6 \times 10^{-5}$ mol dm^{-3}

13 What is the pH of 0.01 mol dm^{-3} solution of HCl which dissociates fully?

$$HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$$

14 For each of the following biological fluids, calculate the pH from the given concentration of H^+ or OH^- ions.

(a) bile: $[OH^-] = 8 \times 10^{-8}$ mol dm^{-3}
(b) gastric juice: $[H^+] = 10^{-2}$ mol dm^{-3}
(c) urine: $[OH^-] = 6 \times 10^{-10}$ mol dm^{-3}

15 A solution of sodium hydroxide is prepared by adding distilled water to 6.0 g NaOH to make 1.0 dm^3 of solution. What is the pH of this solution? Assume that NaOH dissociates completely in solution:

$$NaOH(aq) \rightarrow Na^+(aq) + OH^-(aq)$$

8.4 Strong and weak acids and bases

Understandings:

- Strong and weak acids and bases differ in the extent of ionization.

  Guidance
  The terms ionization and dissociation can be used interchangeably.

- Strong acids and bases of equal concentrations have higher conductivities than weak acids and bases.

  Guidance
  See section 21 in the data booklet for a list of weak acids and bases.

- A strong acid is a good proton donor and has a weak conjugate base.
- A strong base is a good proton acceptor and has a weak conjugate acid.

Applications and skills:

- Distinction between strong and weak acids and bases in terms of the rates of their reactions with metals, metal oxides, metal hydroxides, metal hydrogen carbonates, and metal carbonates and their electrical conductivities for solutions of equal concentrations.

The strength of an acid or base depends on its extent of ionization

We have seen that the reactions of acids and bases are dependent on the fact that they dissociate in solution, acids to produce H^+ ions and bases to produce OH^- ions. As we will see here, the extent of this dissociation is what defines the strength of an acid or a base.

Consider the acid dissociation reaction:

$$HA(aq) + H_2O(l) \rightleftharpoons A^-(aq) + H_3O^+(aq)$$

If this acid dissociates fully, it will exist entirely as ions in solution. It is said to be a **strong acid**. For example, hydrochloric acid, HCl, is a strong acid.

The reaction is written without the equilibrium sign.

$$HCl(aq) + H_2O(l) \rightarrow H_3O^+(aq) + Cl^-(aq)$$

If, on the other hand, the acid dissociates only partially, it produces an equilibrium mixture in which the undissociated form dominates. It is said to be a **weak acid**. For example, ethanoic acid, CH_3COOH, is a weak acid.

$$CH_3COOH(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + CH_3COO^-(aq)$$

Here it is essential to use the equilibrium sign for its dissociation reaction.

Strong acids are good proton donors. As their dissociation reactions go to completion, their conjugate bases are not readily able to accept a proton. For example, HCl reacts to form the conjugate base Cl^-, which shows virtually no basic properties.

$HCl(aq)$	+	$H_2O(l)$	$\rightarrow$	$H_3O^+(aq)$	+	$Cl^-(aq)$
strong acid		base		conjugate acid		conjugate base

Weak acids are poor proton donors. As their dissociation reactions are equilibria which lie to the left, in favour of reactants, their conjugate bases are readily able to accept a proton. For example, CH_3COOH reacts to form the conjugate base CH_3COO^-, which is a stronger base than Cl^-.

$CH_3COOH(aq)$	+	$H_2O(l)$	$\rightleftharpoons$	$H_3O^+(aq)$	+	$CH_3COO^-(aq)$
weak acid		base		conjugate acid		conjugate base

So acid dissociation reactions favour the production of the weaker conjugate.

In a similar way, bases can be described as strong or weak on the basis of the extent of their ionization. For example, NaOH is a **strong base** because it ionizes fully.

$$NaOH(aq) \rightarrow Na^+(aq) + OH^-(aq)$$

Its dissociation is written without equilibrium signs.

(Note that it is the OH^- ions that show Brønsted–Lowry base behaviour by accepting protons.)

On the other hand, NH_3 is a **weak base** as it ionizes only partially, so its equilibrium lies to the left and the concentration of ions is low.

$$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$$

Strong bases are good proton acceptors; they react to form conjugates that do not show acidic properties. Weak bases are poor proton acceptors; they react to form conjugates with stronger acidic properties than the conjugates of strong bases. Base ionization reactions favour the production of the weaker conjugate.

In section 18.2 (page 366), we will learn how we can quantify the strength of an acid or base in terms of the equilibrium constant for its dissociation reaction, and look in more detail at the relationship between the strength of acid–base conjugate pairs.

The strength of an acid or base is therefore a measure of how readily it dissociates in aqueous solution. This is an inherent property of a particular acid or base, dependent on its bonding. Do not confuse acid or basic *strength* with its *concentration*, which is a variable depending on the number of moles per unit volume, according to how much solute has been added to the water. Note, for example, it is possible for an acid or base

Strong acids and strong bases ionize almost completely in solution; weak acids and weak bases ionize only partially in solution.

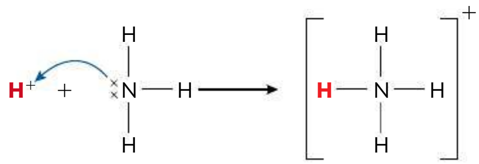
In writing the ionization reactions of weak acids and bases, it is essential to use the equilibrium sign.

Applications and skills:

- Application of Lewis acid–base theory to inorganic and organic chemistry to identify the role of the reacting species.

Lewis theory focuses on electron pairs

Gilbert Lewis, whose name famously belongs to electron dot structures for representing covalent bonding (Chapter 4), used such structures in interpreting Brønsted–Lowry theory. Realizing that the base must have a lone pair of electrons, he reasoned that the entire reaction could be viewed in terms of the electron pair rather than in terms of proton transfer. For example, the reaction previously described in which ammonia acts as a base can be represented as follows:



The curly arrow (shown in blue) is a convention used to show donation of a pair of electrons. H^+ is acting as an electron pair acceptor and the nitrogen atom in ammonia is acting as an electron pair donor. From such thinking Lewis developed a new, broader definition of acids and bases.

A Lewis acid is lone pair acceptor. A Lewis base is a lone pair donor.

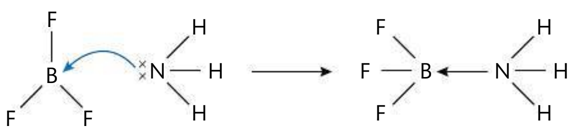
Make sure that when you are describing Lewis acid–base behaviour you refer to donation and acceptance of an electron *pair*. If you omit the word pair, you would be describing a redox reaction which is entirely different.

- A Lewis acid is a lone pair acceptor.
- A Lewis base is a lone pair donor.

Lewis bases and Brønsted–Lowry bases are therefore the same group of compounds: by either definition they are species that must have a lone pair of electrons.

In the case of acids, however, the Lewis definition is broader than the Brønsted–Lowry theory: no longer restricted just to H^+, an acid by Lewis definition is any species capable of accepting a lone pair of electrons. Of course this *includes* H^+ with its vacant orbital (so all Brønsted–Lowry acids *are* Lewis acids) – but it will also include molecules that have an incomplete valence shell. Lewis acid–base reactions result in the formation of a covalent bond, which will always be a **coordinate bond** because both the electrons come from the base.

Here is another example:



BF_3 has an incomplete octet so is able to act as a Lewis acid and accept a pair of electrons; NH_3 acts as a Lewis base, donating its lone pair of electrons. The arrow on the covalent bond denotes the fact that it is a coordinate bond with both electrons donated from the nitrogen.

Other good examples of Lewis acid–base reactions are found in the chemistry of the transition elements. As we learned in Chapter 3, these metals in the middle of the Periodic Table often form ions with vacant orbitals in their d subshell. So they are able to act as Lewis acids and accept lone pairs of electrons when they bond with ligands to form complex ions.

Ligands, as donors of lone pairs, are therefore acting as Lewis bases.

For example, Cu^{2+} in aqueous solution reacts as follows:

$$Cu^{2+}(aq) + 6H_2O(l) \rightarrow [Cu(H_2O)_6]^{2+}(aq)$$

Cu^{2+} is a Lewis acid and H_2O is a Lewis base.

Typical ligands found in complex ions include H_2O, CN^-, and NH_3. Note that these all possess lone pairs of electrons, the defining feature of their Lewis base properties.

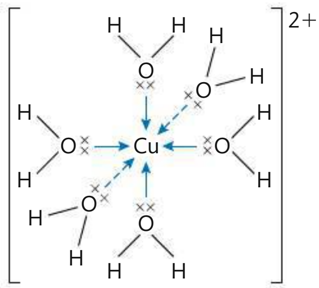


Figure 8.6 In the complex ion$[Cu(H_2O)_6]^{2+}$, Cu^{2+} has acted as a Lewis acid and the H_2O ligands as Lewis bases. The bonds within the complex ion are coordinate bonds, as indicated by the arrows. For clarity only one lone pair (out of two) is shown on each O atom in H_2O.

Copper ions (Cu^{2+}) forming different complex ions with distinct colours. From left to right the ligands are H_2O, Cl^-, NH_3, and the organic group EDTA. The Cu^{2+} ion has acted as the Lewis acid, the ligands as Lewis bases.

Nucleophiles and electrophiles

At about the same time as Lewis developed his acid–base theory, alternate language was being introduced to describe the behaviour of reactants with respect to electron pairs.

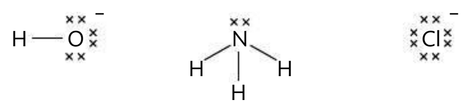
A **nucleophile** ('likes nucleus') is an electron-rich species that donates a lone pair to form a new covalent bond in a reaction.

An **electrophile** ('likes electrons') is an electron-deficient species that accepts a lone pair from another reactant to form a new covalent bond.

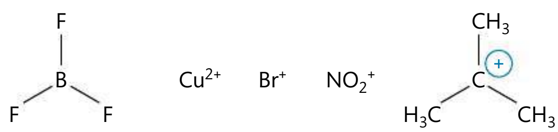
These terms are often used to describe reactions in terms of electron-rich nucleophiles attacking electron-deficient electrophiles, and are depicted using curly arrows to show electron movements. Several examples from organic chemistry are discussed in Chapter 10.

Clearly these terms are the same as those derived by Lewis. In other words, a nucleophile is a Lewis base and an electrophile is a Lewis acid.

- Examples of nucleophiles / Lewis bases:



- Examples of electrophiles / Lewis acids:

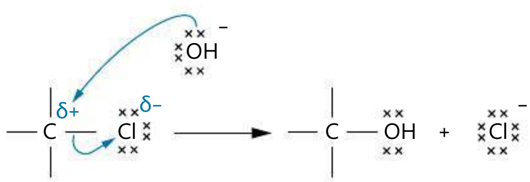


A nucleophile is a Lewis base and an electrophile is a Lewis acid.

TOK

Different approaches and vocabulary can sometimes be used to explain the same phenomenon. Does the use of language as a way of knowing help us to judge these competing approaches?

The reaction below shows the hydroxide ion, OH^-, acting as a nucleophile on an organic molecule known as a halogenoalkane. This reaction is discussed in detail on page 498. We can see that OH^- is a Lewis base as it is donating a lone pair of electrons.



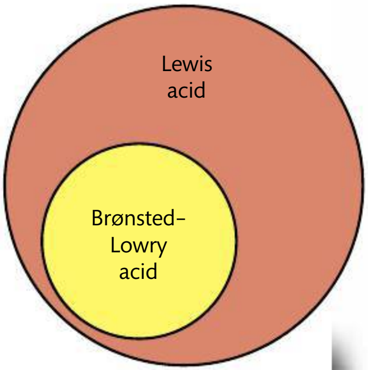


Figure 8.7 Relationship between Brønsted–Lowry acids and Lewis acids.

Comparison of Brønsted–Lowry and Lewis theories of acids and bases

Theory	Definition of acid	Definition of base
Brønsted–Lowry	proton donor	proton acceptor
Lewis	electron pair acceptor	electron pair donor

- Although all Brønsted–Lowry acids are Lewis acids, not all Lewis acids are Brønsted–Lowry acids. The term Lewis acid is usually reserved for those species which can *only* be described by Lewis theory, that is those that do not release H^+.
- Many reactions cannot be described as Brønsted–Lowry acid–base reactions, but do qualify as Lewis acid–base reactions. These are reactions where no transfer of H^+ occurs.

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Lewis acid–base theory is compatible with Brønsted–Lowry theory, it does not falsify it. By changing the perspective of acid–base reactions from the proton to the lone pair of electrons, it opens up a wider field of application. The proton no longer has a central role, it is just one example of a lone pair acceptor. Lewis acids include a wide range of both organic and inorganic species that are not recognized as acids in Brønsted–Lowry theory. Yet, in many cases, as we will see in the following sections, Brønsted–Lowry theory is sufficient and could be considered a more useful theory for the description of acid–base reactions, especially those occurring in aqueous solution.

Exercises

19 For each of the following reactions identify the Lewis acid and the Lewis base.

(a) $4NH_3(aq) + Zn^{2+}(aq) \rightarrow [Zn(NH_3)_4]^{2+}(aq)$
(b) $2Cl^-(aq) + BeCl_2(aq) \rightarrow [BeCl_4]^{2-}(aq)$
(c) $Mg^{2+}(aq) + 6H_2O(l) \rightarrow [Mg(H_2O)_6]^{2+}(aq)$

20 Which of the following could not act as a ligand in a complex ion with a transition metal?

A Cl^- **B** NCl_3 **C** PCl_3 **D** CH_4

21 Which of the following reactions represents an acid–base reaction according to Lewis theory but not according to Brønsted–Lowry theory?

A $NH_3(aq) + HCl(aq) \rightleftharpoons NH_4Cl(aq)$
B $2H_2O(l) \rightleftharpoons H_3O^+(aq) + OH^-(aq)$
C $Cu^{2+}(aq) + 4NH_3(aq) \rightleftharpoons [Cu(NH_3)_4]^{2+}(aq)$
D $BaO(s) + H_2O(l) \rightleftharpoons Ba^{2+}(aq) + 2OH^-(aq)$

18.2 Calculations involving acids and bases

Understandings:

- The expression for the dissociation constant of a weak acid (K_a) and a weak base (K_b).
- For a conjugate acid–base pair, $K_a \times K_b = K_w$.

Guidance

- *The value K_w depends on the temperature.*
- *Only examples involving the transfer of one proton will be assessed.*
- *Calculations of pH at temperatures other than 298 K can be assessed.*

- The relationship between K_a and pK_a is $pK_a = -\log K_a$ and between K_b and pK_b is $pK_b = -\log K_b$.

Applications and skills:

- Solution of problems involving $[H^+(aq)]$, $[OH^-(aq)]$, pH, pOH, K_a, pK_a, K_b, and pK_b.

  Guidance
  - *Students should state when approximations are used in equilibrium calculations.*
  - *The use of quadratic equations will not be assessed.*

- Discussion of the relative strengths of acids and bases using values of K_a, pK_a, K_b, and pK_b.

  Guidance

  The calculation of pH in buffer solutions will only be assessed in options B.7 and D.4.

We have learned that acids and bases differ in their strength according to the equilibrium position of their ionization reactions. They can also be prepared in different concentrations of aqueous solution according to the ratio of acid or base to water used. Both these factors, strength and concentration, influence the pH of a solution. In this section we will learn how to quantify these relationships.

K_w is temperature dependent

The ionic product constant of water, K_w, was derived on pages 358–359.

$$K_w = [H^+]\,[OH^-] = 1.00 \times 10^{-14} \text{ at } 298\text{ K}$$

As K_w is an equilibrium constant, its value must be temperature dependent. The reaction for the dissociation of water is endothermic (it involves bond breaking) and so, as we learned in Chapter 7, an increase in temperature will shift the equilibrium to the right and increase the value of K_w. This represents an increase in the concentrations of $H^+(aq)$ and $OH^-(aq)$, and so a decrease in pH. Conversely, a reduction in temperature, through its effect on shifting the equilibrium to the left, decreases the value of K_w, representing lower ion concentrations and so higher pH values. Some data to illustrate these trends are given in the table below.

Temperature / °C	K_w	$[H^+]$ in pure water ($\sqrt{K_w}$)	pH of pure water ($-\log_{10}[H^+]$)
0	1.5×10^{-15}	0.39×10^{-7}	7.47
10	3.0×10^{-15}	0.55×10^{-7}	7.27
20	6.8×10^{-15}	0.82×10^{-7}	7.08
25	1.0×10^{-14}	1.00×10^{-7}	7.00
30	1.5×10^{-14}	1.22×10^{-7}	6.92
40	3.0×10^{-14}	1.73×10^{-7}	6.77
50	5.5×10^{-14}	2.35×10^{-7}	6.63

In other words, the pH of pure water is 7.00 only when the temperature is 298 K. Note that at temperatures above and below this, despite changes in the pH value, water is still a neutral substance as its $H^+(aq)$ concentration is equal to its $OH^-(aq)$ concentration. It does not become acidic or basic as we heat it and cool it respectively!

The temperature dependence of K_w means that the temperature should always be stated alongside pH measurements.

TOK

Because many people are so familiar with the value 7.00 as the pH of water, it is often more difficult to convince them that water with a pH of greater or less than this is still neutral. Can you think of other examples where entrenched prior knowledge might hinder a fuller understanding of new knowledge? Are we likely to misinterpret experimental data when it does not fit with our expectations based on prior knowledge?

pH and pOH scales are inter-related

In section 8.3 we learned that the pH scale was introduced in order to simplify the expression of the H^+ concentration in a solution, and in particular it helps us to compare different solutions in terms of their H^+ content. The same rationale can be applied to the OH^- ions. Like H^+ ions, OH^- ions are often present in low concentrations in solutions and so have negative exponents when expressed as mol dm^{-3} that can be awkward to work with. The parallel scale, known as the **pOH scale**, is therefore used to describe the OH^- content of solutions.

- $pOH = -\log_{10}[OH^-]$;
- $[OH^-] = 10^{-pOH}$
- $pH = -\log_{10}[H^+]$;
- $[H^+] = 10^{-pH}$

As explained on page 356, the logarithmic nature of these scales means that a change of one unit in pH or pOH represents a 10× change in $[H^+]$ or $[OH^-]$ respectively. The scales are inverse, so the higher the H^+ or OH^- concentration, the smaller the pH or pOH value. These values are usually positive and have no units.

In all aqueous solutions at 298 K: pH + pOH = 14

From the relationship $[H^+][OH^-] = K_w = 1.00 \times 10^{-14}$ at 298 K, it follows that

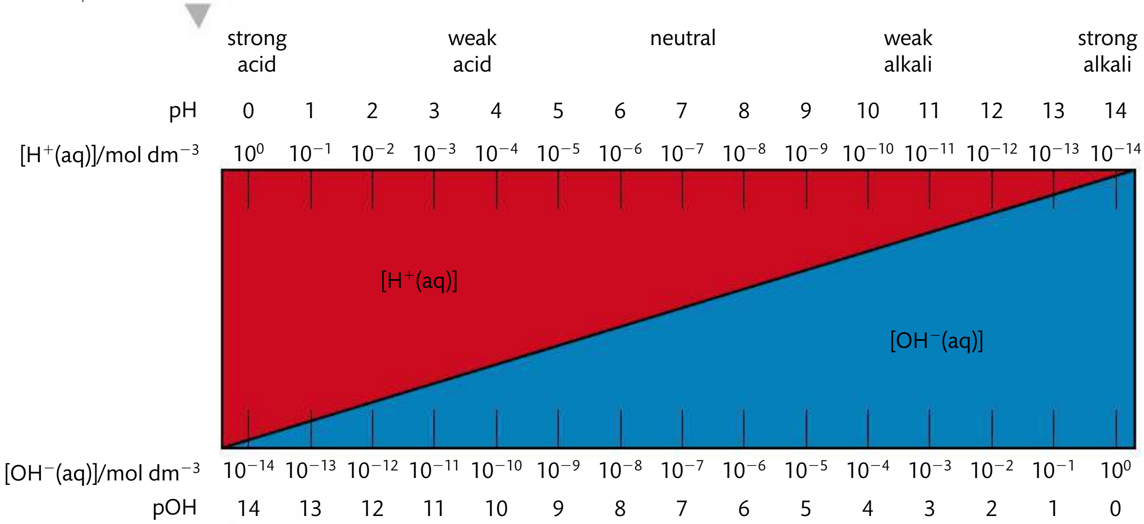
$$10^{-pH} \times 10^{-pOH} = 1.00 \times 10^{-14} \text{ at 298 K}$$

By taking the negative logarithm to base 10 of both sides, we get

$$pH + pOH = 14.00 \text{ at 298 K}$$

The relationships between pH, $[H^+]$, pOH, and $[OH^-]$ are shown in Figure 8.8.

Figure 8.8 Relationship between $[H^+]$, $[OH^-]$, pH, and pOH at 298 K.



In the same way as the negative logarithms to base 10 of H^+ and OH^- are known as pH and pOH respectively, the same terminology can be applied to K_w to derive $\mathbf{p}\boldsymbol{K}_\mathbf{w}$.

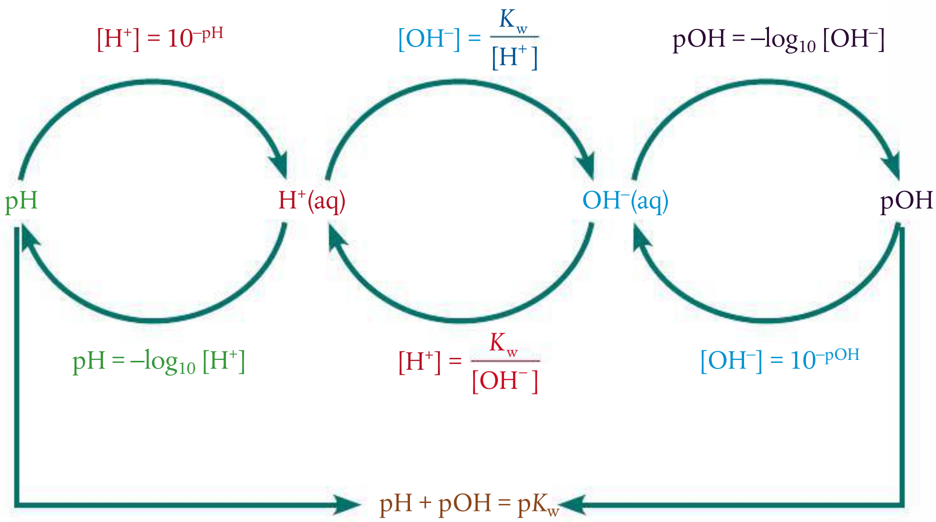
- $pK_w = -\log_{10}(K_w)$
- $K_w = 10^{-pK_w}$

pH + pOH = pK_w

So we can rewrite the expression above in a form that will apply to all temperatures:

$$pH + pOH = pK_w$$

Summary of the relationships between [H⁺], [OH⁻], pH, and pOH



Converting H^+ and OH^- into pH and pOH

Many calculations on acids and bases involve inter-conversions of $[H^+]$ and pH, as we saw in section 8.3, and we can extend these now to include $[OH^-]$ and pOH.

Worked example

Lemon juice has a pH of 2.90 at 25 °C. Calculate its $[H^+]$, $[OH^-]$, and pOH.

Solution

$$[H^+] = 10^{-pH} = 10^{-2.90} = 1.3 \times 10^{-3}\ \text{mol dm}^{-3}$$

$pH + pOH = 14.00$ at 298 K, so $pOH = 14.00 - 2.90 = 11.10$

$[OH^-] = 10^{-pOH} = 10^{-11.10} = 7.7 \times 10^{-12}\ \text{mol dm}^{-3}$

or $K_w = [H^+][OH^-]$ so $1.00 \times 10^{-14} = (1.3 \times 10^{-3}) \times [OH^-]$

$[OH^-] = 7.7 \times 10^{-12}\ \text{mol dm}^{-3}$

Household products containing acids, alongside sulfuric acid. Malt vinegar contains ethanoic acid, orange juice and lemons contain citric acid, and automobile batteries usually contain sulfuric acid.

Strong acids and bases: pH and pOH can be deduced from their concentrations

We assume full dissociation for strong acids and bases. Because of this we can deduce the ion concentrations and so calculate the pH or pOH directly from the initial concentration of the solution. Note that the pH and pOH are derived from the *equilibrium* concentrations of H^+ and OH^-. In these examples we will use the same notation as in Chapter 7, black for data that are given in the questions and blue for derived data.

Worked example

Calculate the pH of the following at 298 K:

(a) 0.10 mol dm^{-3} NaOH(aq)
(b) 0.15 mol dm^{-3} H_2SO_4(aq)

Solution

(a)	NaOH(aq) →	Na^+(aq) +	OH^-(aq)
initial (mol dm^{-3})	0.10		
equilibrium (mol dm^{-3})			0.10

pOH = $-\log_{10}$ (0.10) = 1.0, therefore pH = 13.0

(b)	H_2SO_4(aq) →	$2H^+$(aq) +	SO_4^{2-}(aq)
initial (mol dm^{-3})	0.15		
equilibrium (mol dm^{-3})		0.30	

pH = $-\log_{10}$ (0.30) = 0.52

Exercises

22 At the temperature of the human body, 37 °C, the value of $K_w = 2.4 \times 10^{-14}$.
Calculate [H^+], [OH^-], pH, pOH, and pK_w of water at this temperature. Is it acidic, basic, or neutral?

23 The pH of a sample of milk is 6.77 at 298 K. Calculate its pOH, [H^+], and [OH^-]. Deduce whether it is acidic, basic, or neutral.

24 Calculate the pH of the following solutions:
(a) 0.40 mol dm^{-3} HCl
(b) 3.7×10^{-4} mol dm^{-3} KOH
(c) 5.0×10^{-5} mol dm^{-3} $Ba(OH)_2$

25 Which values are correct for a solution of NaOH of concentration 0.010 mol dm^{-3} at 298 K?
A [H^+] = 1.0×10^{-2} mol dm^{-3} and pH = 2.00
B [OH^-] = 1.0×10^{-2} mol dm^{-3} and pH = 12.00
C [H^+] = 1.0×10^{-12} mol dm^{-3} and pOH = 12.00
D [OH^-] = 1.0×10^{-12} mol dm^{-3} and pOH = 2.00

Dissociation constants express the strength of weak acids and bases

Weak acids and bases, unlike strong acids and bases, do not dissociate fully. This means we *cannot* deduce the concentrations of ions in their solutions from the initial concentrations, as the ion concentrations will depend on the extent of dissociation that has occurred. So we need some means of quantifying the extent of dissociation – and the process takes us back to equilibrium considerations.

The dissociation reactions of weak acids and weak bases can be represented as equilibrium expressions, each with their own equilibrium constant. The value of this constant will convey information about the position of equilibrium, and therefore on the extent of dissociation of the acid or base.

Consider the generic weak acid HA dissociating in water:

$$HA(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + A^-(aq)$$

$$K_c = \frac{[H_3O^+][A^-]}{[HA][H_2O]}$$

Given that the concentration of water is considered to be a constant, we can combine this with K_c to produce a modified equilibrium constant known as $\boldsymbol{K_a}$.

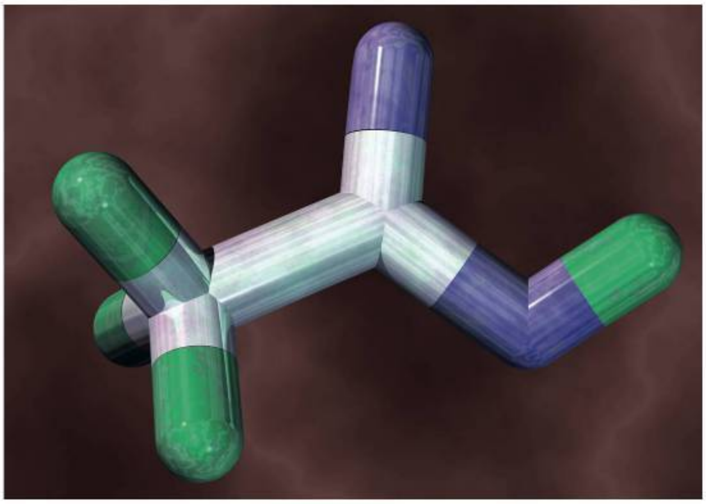
$$K_c[H_2O] = \frac{[H_3O^+][A^-]}{[HA]}$$

$$\text{Therefore } K_a = \frac{[H_3O^+][A^-]}{[HA]}$$

K_a is known as the **acid dissociation constant**. It will have a fixed value for a particular acid at a specified temperature.

Note that the term **acid dissociation constant** is sometimes used interchangeably with the term **acid ionization constant**. You should be comfortable to recognize either terminology in textbooks and questions.

As the value of K_a depends on the position of the equilibrium of acid dissociation, it gives a direct measure of the strength of an acid. *The higher the value of K_a at a particular temperature, the greater the dissociation, and so the stronger the acid.* Note that because K_a is an equilibrium constant, its value does not depend on the concentration of the acid or on the presence of other ions. We will return to this point in our study of buffer solutions in section 18.3 (page 378).



Computer model of a molecule of ethanoic acid, CH_3COOH. The atoms (tubes) are colour coded: carbon is white, hydrogen is green, and oxygen is purple. Ethanoic acid is a weak acid, and is used in many of the examples in this chapter. It is used in the production of plastics and also as a preservative.

Similarly, we can consider the ionization of a base using the generic weak base B.

$$B(aq) + H_2O(l) \rightleftharpoons BH^+(aq) + OH^-(aq)$$

$$K_c = \frac{[BH^+][OH^-]}{[B][H_2O]}$$

Again we can combine the constants to give a modified equilibrium constant $\boldsymbol{K_b}$.

$$K_c[H_2O] = \frac{[BH^+][OH^-]}{[B]}$$

$$\text{Therefore } K_b = \frac{[BH^+][OH^-]}{[B]}$$

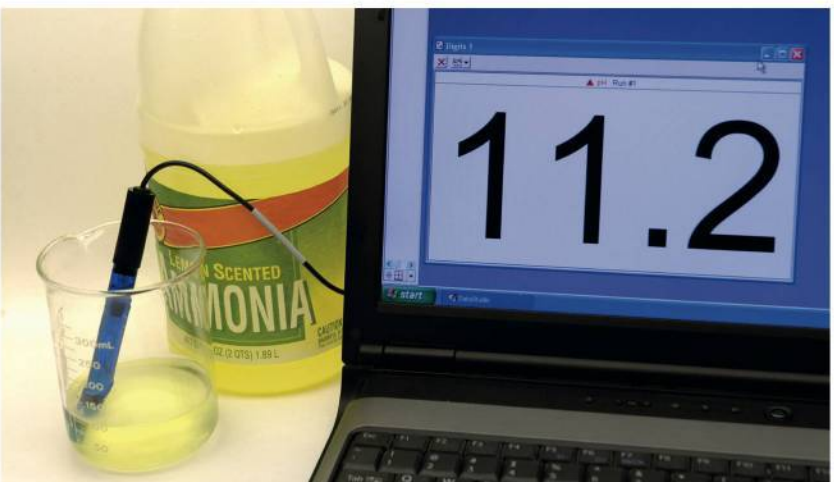
K_b is known as the **base dissociation constant**. It will have a fixed value for a particular base at a specified temperature.

As with K_a, the value of K_b relates to the position of the equilibrium and so in this case to the strength of the base. *The higher the value of K_b at a particular temperature, the greater the ionization and so the stronger the base.*

$$K_a = \frac{[H_3O^+][A^-]}{[HA]}$$

$$K_b = \frac{[BH^+][OH^-]}{[B]}$$

The values of K_a and K_b are constant for a specific acid and base respectively, at a specified temperature. Their values give a measure of the strength of the acid or base.



A pH probe interfaced to a computer showing ammonia solution to have a pH of 11.2. It is a weak base, used in many of the examples in this chapter.

Worked example

Write the expressions for K_a and K_b for the following acid and base.

(a) $CH_3COOH(aq)$

(b) $NH_3(aq)$

Solution

First write the equation for the equilibrium reactions – remembering that acids donate H^+ and bases accept H^+.

(a) $CH_3COOH(aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$

$$K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]}$$

(b) $NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$

$$K_b = \frac{[NH_4^+][OH^-]}{[NH_3]}$$

Exercises

26 Write the expressions for K_b for the following:

(a) $C_2H_5NH_2(aq)$
(b) $HSO_4^-(aq)$
(c) $CO_3^{2-}(aq)$

27 Place the following acids in order of increasing strength.

H_3PO_4 $K_a = 7.1 \times 10^{-3}$
HNO_2 $K_a = 7.1 \times 10^{-4}$
H_2SO_3 $K_a = 1.2 \times 10^{-2}$

28 Why do you think we do not usually use the concept of K_a and K_b to describe the strength of strong acids and bases?

Calculations involving K_a and K_b

The values of K_a and K_b enable us to compare the strengths of weak acids and bases, and to calculate ion concentrations present at equilibrium and so the pH and pOH values. Examples of these calculations are given below. Note that many of these questions are approached in a similar manner to those given in Chapter 7. It is assumed here that you are familiar with that work, so it may be useful for you to refresh your memory of section 17.1 before continuing.

The following points are to remind you of some key points and guide you in all the calculations that follow:

- The given concentration of an acid or base is its *initial* concentration – before dissociation occurs.
- The pH (or pOH) of a solution refers to the concentration of H^+ ions (or OH^- ions) at *equilibrium*.
- The concentration values substituted into the expressions for K_a and K_b must be the *equilibrium* values for all reactants and products.
- When the extent of dissociation is very small (very low value for K_a or K_b) it is appropriate to use the approximations:

$$[\text{acid}]_{\text{initial}} \approx [\text{acid}]_{\text{equilibrium}}$$

$$[\text{base}]_{\text{initial}} \approx [\text{base}]_{\text{equilibrium}}$$

Make sure you clearly state when an approximation is made in a calculation, and justify it in terms of the small value of K.

1 Calculation of K_a and K_b from pH and initial concentration

Worked example

Calculate K_a at 298 K for a 0.01mol dm^{-3} solution of ethanoic acid (CH_3COOH). It has a pH of 3.4 at this temperature.

Solution

Write the equation for the dissociation of the acid. Insert the data in three rows: initial, change, and equilibrium. As in Chapter 7, numbers in black are data that were given in the question, numbers in blue have been derived.

From the pH we get the [H^+] at equilibrium:

pH 3.4 $\Rightarrow$ [H^+] = $10^{-3.4}$ = 4.0×10^{-4} mol dm^{-3}

From the stoichiometry of the reaction we know that [H^+] = [CH_3COO^-]

	$CH_3COOH(aq)$	$\rightleftharpoons$	$CH_3COO^-(aq)$	+	$H^+(aq)$
initial (mol dm^{-3})	0.01		0.00		0.00
change (mol dm^{-3})	-4×10^{-4}		$+4 \times 10^{-4}$		$+4 \times 10^{-4}$
equilibrium (mol dm^{-3})	$0.01 - (4 \times 10^{-4})$ ~ 0.01		4×10^{-4}		4×10^{-4}

The approximation $0.01 \approx 0.01 - (4 \times 10^{-4})$ is valid within the precision of this data.

Write the expression for K_a and substitute the equilibrium values.

$$K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]} = \frac{(4\times10^{-4})^2}{0.01} = 1.6\times10^{-5}$$

Worked example

Calculate the K_b for a 0.100 mol dm^{-3} solution of methylamine, CH_3NH_2 at 25 °C . Its pH is 11.80 at this temperature.

Solution

At 25 °C (298 K), pH + pOH = 14.00. Therefore pH 11.80 $\Rightarrow$ pOH = 2.20

[OH^-] = 10^{-pOH} = $10^{-2.20}$ = 6.3×10^{-3}

From the stoichiometry of the reaction [OH^-] = [$CH_3NH_3^+$]

	$CH_3NH_2(aq)$	+	$H_2O(l)$	$\rightleftharpoons$	$CH_3NH_3^+(aq)$	+	$OH^-(aq)$
initial (mol dm^{-3})	0.100				0.000		0.000
change (mol dm^{-3})	−0.00630				+0.00630		+0.00630
equilibrium (mol dm^{-3})	0.0937				0.00630		0.00630

$$K_b = \frac{[CH_3NH_3^+][OH^-]}{[CH_3NH_2]} = \frac{(0.00630)^2}{0.0937} = 4.22\times10^{-4}$$

2 Calculation of $[H^+]$ and pH, $[OH^-]$ and pOH from K_a and K_b

Worked example

A 0.75 mol dm^{-3} solution of ethanoic acid has a value for $K_a = 1.8 \times 10^{-5}$ at a specified temperature. What is its pH at this temperature?

Solution

To calculate pH we need to know $[H^+]$ at equilibrium, and therefore the amount of dissociation of the acid that has occurred: this is the 'change' amount in the reaction.

So let the change in concentration of $CH_3COOH = -x$

Therefore change in concentration of CH_3COO^- and $H^+ = +x$

	$CH_3COOH(aq)$	$\rightleftharpoons$	$CH_3COO^-(aq)$	+	$H^+(aq)$
initial (mol dm^{-3})	0.75		0.00		0.00
change (mol dm^{-3})	$-x$		$+x$		$+x$
equilibrium (mol dm^{-3})	$0.75 - x$ ~ 0.75		x		x

As K_a is very small, x, the amount of dissociation, is also extremely small and it is valid to approximate $[CH_3COOH]_{initial} \approx [CH_3COOH]_{equilibrium}$.

$$K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]} = \frac{x^2}{0.75} = 1.8 \times 10^{-5}$$

Therefore $x = \sqrt{1.8 \times 10^{-5} \times 0.75} = 3.7 \times 10^{-3}$

$[H^+] = 3.7 \times 10^{-3} \Rightarrow pH = 2.4$

Worked example

A 0.20 mol dm^{-3} aqueous solution of ammonia has K_b of 1.8×10^{-5} at 298 K . What is its pH?

Solution

Let the change in concentration of $NH_3 = -x$

Therefore change in concentration of NH_4^+ and $OH^- = +x$

	$NH_3(aq)$	+	$H_2O(l)$	$\rightleftharpoons$	$NH_4^+(aq)$	+	$OH^-(aq)$
initial (mol dm^{-3})	0.20				0.00		0.00
change (mol dm^{-3})	$-x$				$+x$		$+x$
equilibrium (mol dm^{-3})	$0.20 - x$ ≈ 0.20				x		x

As K_b is very small, x the amount of dissociation is also extremely small, and so it is valid to approximate $[NH_3]_{initial} \approx [NH_3]_{equilibrium}$

$$K_b = \frac{[NH_4^+][OH^-]}{[NH_3]} = \frac{x^2}{0.20} = 1.8 \times 10^{-5}$$

Therefore $x = \sqrt{1.8 \times 10^{-5} \times 0.20} = 1.9 \times 10^{-3}$

$[OH^-] = 1.9 \times 10^{-3}$

$pOH = -\log_{10}(1.9 \times 10^{-3}) = 2.72$

Therefore at 298 K, pH = 14.00 – 2.72 = 11.28

Exercises

29 The acid dissociation constant of a weak acid has a value of 1.0×10^{-5} mol dm^{-3}. What is the pH of a 0.1 mol dm^{-3} aqueous solution of HA?

A 2 **B** 3 **C** 5 **D** 6

30 Calculate the K_b of ethylamine, $C_2H_5NH_2$, given that a 0.10 mol dm^{-3} solution has a pH of 11.86.

31 What are the [H^+] and [OH^-] in a 0.10 mol dm^{-3} solution of an acid that has $K_a = 1.0 \times 10^{-7}$?

▲ Stinging nettle plants. The leaves are covered in sharp hairs which when touched inject a painful mixture of chemicals including the weak acid methanoic acid HCOOH into the skin. The stinging sensation can be relieved by rubbing the skin with leaves of a plant such as dock, which has a mildly basic sap and so helps to neutralize the acid.

pK_a and pK_b

We have seen that K_a and K_b values give us a direct measure of the relative strengths of weak acids and bases. But as these values are characteristically very small, they usually involve dealing with numbers with negative exponents, which we have acknowledged before are clumsy to use as the basis for comparisons. They also span a wide range of values. So, in the same way as with the concentrations of H^+ and OH^- ions and K_w, and for the same reason, we can convert K_a and K_b values into their negative logarithms to the base 10, known as **pK_a** and **pK_b**.

- p$K_a = -\log_{10} K_a$;
- $K_a = 10^{-pK_a}$
- p$K_b = -\log_{10} K_b$;
- $K_b = 10^{-pK_b}$

Some examples of K_a and pK_a, K_b and pK_b values are given below, all at 298 K.

Acid	Formula	K_a	pK_a
methanoic	HCOOH	1.8×10^{-4}	3.75
ethanoic	CH_3COOH	1.8×10^{-5}	4.76
propanoic	C_2H_5COOH	1.4×10^{-5}	4.87

Base	Formula	K_b	pK_b
ammonia	NH_3	1.8×10^{-5}	4.75
methylamine	CH_3NH_2	4.6×10^{-4}	3.34
ethylamine	$C_2H_5NH_2$	4.5×10^{-4}	3.35

The following points follow from the table.

1 **pK_a and pK_b numbers are usually positive and have no units.**

Although the derivation of K_a and K_b can be applied to any acid or base, it is really useful mostly for weak acids and bases where the extent of dissociation is small. The fact that these have negative powers means that their pK_a and pK_b values will be positive, as we can see in the examples in the table above.

2 **The relationship between K_a and pK_a and between K_b and pK_b is inverse.**

Stronger acids or bases with higher values for K_a or K_b have lower values for pK_a or pK_b.

STRONG ACID

K_a increases ↑↓ pK_a increases

WEAK ACID

STRONG BASE

K_b increases ↑↓ pK_b increases

WEAK BASE

CHALLENGE YOURSELF

1 In organic acids and bases, increasing the length of the carbon chain *decreases* the acid strength of the –COOH group but *increases* the basic strength of $C_2H_5NH_2$ relative to CH_3NH_2. Can you suggest why this is so? Hint: think about the electron density distributions in the molecules.

pK$_a$ = −log$_{10}$ K$_a$; K$_a$ = 10^{-pK_a}
pK$_b$ = −log$_{10}$ K$_b$; K$_b$ = 10^{-pK_b}

- the larger the pK$_a$, the weaker the acid
- the larger the pK$_b$, the weaker the base

3 **A change of one unit in pK_a or pK_b represents a 10 fold change in the value of K_a or K_b.**

This is because the scale is logarithmic to base 10.

4 **pK_a and pK_b must be quoted at a specified temperature.**

This is because the values are derived from the temperature-dependent constants, K_a and K_b.

Table 21 in the IB data booklet gives pK_a and pK_b values for a range of common weak acids and bases. These are the data commonly quoted to describe acid and base strengths, so in calculation questions you may first need to convert pK_a and pK_b into K_a or K_b values. This is done by taking anti-logarithms, as we did with pH, pOH, and pK_w.

Relationship between K_a and K_b, pK_a and pK_b for a conjugate pair

Consider the K_a and K_b expressions for a conjugate acid–base pair HA and A^-.

$$HA(aq) \rightleftharpoons H^+(aq) + A^-(aq) \qquad K_a = \frac{[H^+][A^-]}{[HA]}$$

$$A^-(aq) + H_2O(l) \rightleftharpoons HA(aq) + OH^-(aq) \qquad K_b = \frac{[HA][OH^-]}{[A^-]}$$

$$K_a \times K_b = \frac{[H^+]\cancel{[A^-]}}{\cancel{[HA]}} \times \frac{\cancel{[HA]}[OH^-]}{\cancel{[A^-]}}$$

Therefore $K_a \times K_b = [H^+] \times [OH^-] = K_w$

By taking negative logarithms of both sides:

pK_a + pK_b = pK_w

At 298 K, $K_w = 1.00 \times 10^{-14}$ so pK_w = 14.00

Therefore pK_a + pK_b = 14.00 at 298 K

This relationship holds for any conjugate acid–base pair in aqueous solution, so from the value of K_a of the acid we can calculate K_b for its conjugate base. It shows that the higher the value of K_a for the acid, the lower the value of K_b for its conjugate base. In other words, stronger acids have weaker conjugate bases, and vice versa. This gives a quantitative basis for the relative strengths of conjugate acid–base pairs, discussed qualitatively on page 361. It makes sense in the context of equilibrium positions. For a conjugate pair such as

$$HA(aq) \rightleftharpoons H^+(aq) + A^-(aq)$$

the weaker the acid HA, the further this equilibrium lies to the left; this means the stronger the base A^-, the greater its tendency to accept the proton. So acids and bases react to form the weaker conjugate, as illustrated in Figure 8.9.

For any conjugate acid–base pair:

- $K_a \times K_b = K_w$
- pK_a + pK_b = pK_w
- pK_a + pK_b = 14 at 298 K